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Equipment detection and recognition in electric power room based on faster R-CNN

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Abstract

Faster RCNN is a popular and very efficient object detection algorithm, it has good performance in image recognition in various fields. In this paper, we applied the Faster RCNN algorithm to the equipment recognition and status detection tasks in the electric power room, then proposed a complete technical implementation scheme. First of all, after site investigation and shooting, we collected a total of 5600 pictures in 100 categories of electrical equipment. Then, we achieved the labeled pictures for all categories by employing an open source software named Labelimg. In the next step, the annotated data was randomly shuffled, 70% of which was used as the training set for model training, 10% of which was used as the validation set for model tuning, and the remaining data was used as the test set for model testing. We get less than 0.01% loss in training step and get 91.3% mAP for all test pictures in testing step. Besides, the test for video received a great performance. The well performance on the test pictures and videos showed the effectiveness and implementability of our technical solutions.

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Keywords: Electric power room, faster r-cnn, objection detection, equipment recognition

1. Background

Electricity, as a major basic service industry related to the national economy and people's livelihood, is an important pillar of our country's industrialization construction ^[1]. With the continuous acceleration of country's modernization process and the continuous development of e-commerce and enterprise informatization, the electric power business has developed rapidly, the automation and intelligence level of the electric power system has also been continuously improved. Electric power information and communication management have great importance in

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the power business. How to scientifically and efficiently complete the tasks of power information and communication management is the key to enhancing the core competitiveness of the power business. At present, the research of power information system management automation is mainly focused on automatic inspection and equipment resource management. The most important thing is to realize the detection of target equipment in the power room.

Deep learning has been applied to various fields now, and its application scenarios covered three main areas: object recognition, target detection, and natural language processing [2]. Based on the research of deep learning target detection algorithms, this paper proposed a Faster R-CNN-based power room equipment detection method. We extracted images for the operating state of the equipment at intervals. These images were trained offline to obtain the corresponding equipment detection model, then online real-time monitoring was realized through the automatic inspection robot system, so as to obtain the status information of the equipment in a timely and effective manner. Besides that, we applied the detection model to computer room inspection and configuration items audit.

This article mainly verifies the feasibility of power equipment identification with deep learning method, which effectively solves the problem of detecting equipment in the power room. As a result of it, we saved human and material resources and realized the automatic operation and maintenance of the electric room.

2. Related works

Object detection task could be divided into object recognition and object positioning, which means not only to identify the category of object but more importantly to find the specific location in the picture. Therefore, object detection models are grouped into two categories. One is called two-stage, which carry out object recognition and positioning separately, its representatives are R-CNN families such as fast R-CNN [3] and faster-RCNN [4]. Two-stage method have lower recognition error but slower speed so that they cannot satisfy the need of real-time detection. For which, another type called one-stage emerged, its typical representatives are Yolo, SSD[5], YoloV2[6], YoloV3[7], etc. One stage method can meet real-time requirements for its fast recognition speed and better accuracy.

R-CNN algorithm proposed in 2014 basically laid the foundation of the two-stage method in the field of object detection. Compared with traditional detection algorithm, R-CNN has achieved a 50% performance improvement. Object detection model with VGG-16 network can achieve 66% accuracy on the voc2007 data set. However, R-CNN algorithm also has shortcomings such as slow speed and large memory usage. Thence, fast R-CNN was proposed by Microsoft in 2015, it adds the ROI pooling layer before fully connected layer so that the image does not need to be cropped. Moreover, Faster R-CNN proposes RPN network to obtain candidate frames, which gets rid of the selective search algorithm and only requires one convolutional layer operation, so it greatly improves the recognition speed. Faster R-CNN uses the resNet101 model as the convolutional layer, which can achieve 83.8% accuracy on the voc2012 data set, which exceeds Yolo, SSD and YoloV2. The biggest problem for Faster R-CNN is the slow speed, it can only process 5 frames per second, which cannot meet the real-time requirements.

On account of slow perform speed for the two-stage method of object detection, Yolo creatively proposed one-stage method, it directly returns to the position of the bounding box and the category to which the bounding box belongs in the output layer. In this way, Yolo can achieve a calculation speed of 45 frames per second and fully meet the real-time requirements. However, its mAP and recall score are not satisfactory. SSD combines the advantages and disadvantages of Yolo. For an input image with 300x300 pixels, tested on the voc2007 data set, it can reach 58 frames per second (Titan X GPU) and 72.1% mAP. Aiming at Yolo's weak points such as low accuracy, easy to mistakenly identified, and poor effect on objects that uncommon category, yoloV2 is proposed. It mainly uses Yolo's network structure, combines with the characteristics of SSD, and has made some optimizations and improvements on its basis. YOLOv3 and YOLOv4 are more about integrating other people's work into YOLO under the premise of ensuring speed to improve accuracy.

Considering that the research content of this article requires relatively high recognition accuracy and not very strict time requirements, Faster R-CNN algorithm was applied to the object detection task in power equipment room. This article uses Faster R-CNN algorithm to do experiments on pictures with different shooting angles, partial obstructions, and different lighting conditions, and finally got good detection results.

3. Equipment detection and recognition in electric power room

Information computer room is an important part of power company assets, and computer room inspection is a key link in computer room operation and maintenance. Manual inspection has some shortcomings such as timeliness, poor reliability, more labor costs, greater risks in the inspection process, and low inspection efficiency. The actual application of intelligent inspection robots in the data center will gradually replace the operator to complete the daily inspection of the equipment in the computer room. Here, whether to rely on computer algorithms to achieve more accurate computer room equipment detection and identification will be the primary task.

3.1. Dataset

We have taken multiple sets of videos and images with high-definition cameras through professionals, and extracted video frames of the equipment's operating state from these videos, including normal operating images and faulty images, which used as the sample set of the Faster R-CNN algorithm.

3.2. Algorithm flow

Since there are many types of equipment in the power room and a huge amount of image data collected, it is difficult to check one by one by the staff. This paper applies the deep learning algorithm to the detection problem of the equipment in the power room. We constructed Faster R-CNN algorithm for detection and recognition model on the MXNet platform. First of all, we configured the operating environment of the MXNet platform, and built Faster R-CNN algorithm on the platform. Then labeled and stored the selected sample set according to the PASCAL VOC 2007 data set standard. The parameters are fine-tuned based on the VGG16 pre-training model, so that the final model is suitable for the fault detection and identification of power room equipment. The algorithm flow is shown in Figure 1.

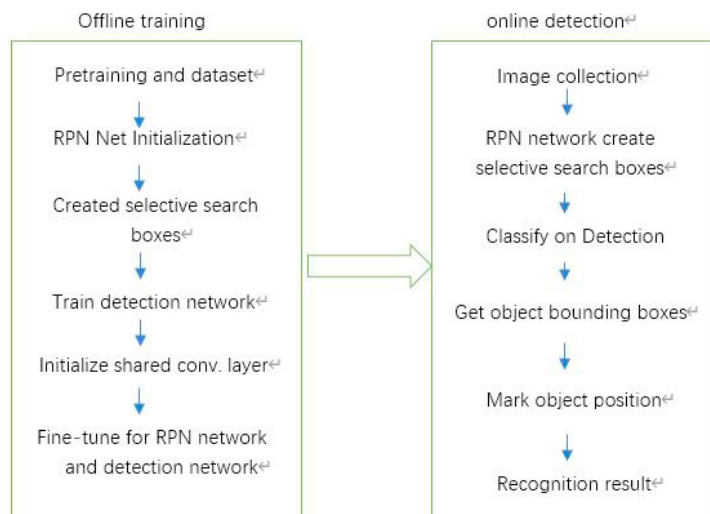


Fig. 1. Algorithm workflow

Simply put, the process of training a model is described as follows: firstly, initialize a model that can hardly do anything, then grab some labeled data sets, modify the model so that it can perform tasks more accurately on the grabbed data sets, and keep repeating above steps until the model perform well. Faster R-CNN algorithm uses pre-labeled VOC data sets for model training, and extracts fine-tuned models by analyzing these labeled data, so that the final model is suitable for equipment failure Detection and identification tasks.

The Faster R-CNN algorithm uses two networks, RPN and Fast R-CNN. Because it is more cumbersome to directly use the BP (Back Propagation) algorithm for training, when training the model, a ResNet50 network is used to extract pictures. The characteristics of the RPN network and the Fast R-CNN network are used for cross-training. Due to the sharing of convolutional features during training, it not only reduces the amount of parameters during training, but also improves training efficiency, and basically achieves quasi-real-time detection performance.

4. Experiments and Analysis

This experiment is mainly to verify the effectiveness and superiority of the Faster R-CNN algorithm in the power room equipment task. Faster R-CNN algorithm can detect the information and status of power room equipment timely and effectively, it is suitable for computer room inspections and resources audit projects, and can effectively improve the efficiency of the automation of the operation and maintenance in power equipment room.

4.1. Dataset

The sample data was collected by professional shooting, including video and image files. We selected 5600 device images as sample set, in total 100 categories. The entire data set was labeled manually and be saved to the format of the PASAL VOC2007 sample set, which shown in figure 4.1. Then we randomly divided the data set into a training set and a data set (quantity ratio 9:1).

4.2. Performance Evaluation

The indicator to measure the detection accuracy in object detection is mAP (mean Average Precision), which refers to the Mean of the average accuracy of multiple categories. AP refers to the average accuracy of a single category, which measures whether the model's recognition in that category is good or bad, while mAP measures the recognition effect of the model on all categories. The value of mAP is between 0-1, larger mAP means better the recognition accuracy.

4.3. Experiments analysis

The environments the experiments used is described as follows: Server system is Microsoft Windows 10 64-bit with 32 GB memory, CPU card is Intel Core i7-8700 with 3.2 GHz, and the graphics card is NVIDIA GeForce GTX 1080 Ti with 11 GB memory. We Build deep learning framework MXNet on the server, and trained Faster R-CNN model with resnet50 backbone.

4.3.1. Comparison of performance for different parameters of Faster R-CNN

In the process of training Faster R-CNN, in order to have better performance for the gradient descent method, the value of the learning rate needs to be set in a suitable range. Too large learning rate leads to unstable learning, too Small values lead to extremely long training time. The selection of the learning rate, attenuation coefficient and momentum parameters of the training model directly affects the training speed and results of the final detection model. This article selects some of the more commonly used learning rates and attenuation coefficients as candidate values. As shown in Table 1. The attenuation coefficient is determined to be 0.1, the selected values for the learning rate are 0.1, 0.01, 0.001, and the selected values for the momentum parameter are 0.3, 0.5 and 0.9. With the attenuation coefficient unchanged, the learning rate is first determined, and then the momentum is determined. Among them, when the learning rate is 0.1, the training fails to converge, which may be the reason that the initial value of the learning rate is too large. As can be seen from the table, the final attenuation coefficient is determined to be 0.1, the initial learning rate is 0.001, and the momentum parameter size is 0.9.

Table 1. comparison of performance for different parameters of Faster R-CNN

Order	attenuation coefficient	Learning rate	Momentum parameters	mAP
1	0.1	0.1	-	/
2	0.1	0.01	-	0.8392
3	0.1	0.001	-	0.8214
4	0.1	0.001	0.3	0.8513
5	0.1	0.001	0.5	0.8725
6	0.1	0.001	0.9	0.9024

Figure 2 described the loss function value during training step for Faster R-CNN, as it is shown, when the iterating to 6000, the loss function value tends to be flat.

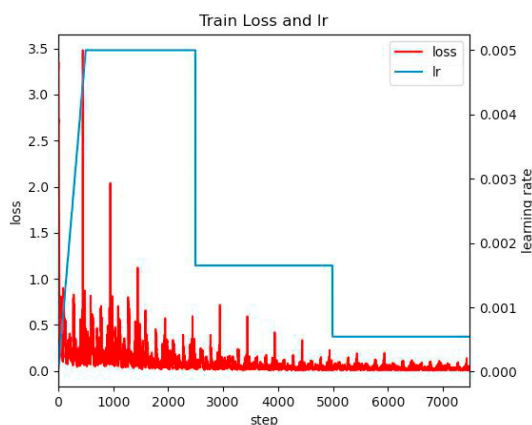


Fig. 2. Train loss graph of Faster R-CNN

4.3.2. Comparison of performance for different RPN network

Since the RPN network is the core network of Faster R-CNN, it greatly improves the efficiency of obtaining candidate suggestion boxes. The Fast R-CNN algorithm obtains candidate suggestion boxes through a selective search method, and applies these two algorithms to detection equipment, Table 2 shows the experimental results.

Table 2. Comparison of performance for different RPN network of Faster R-CNN

Model	Fast R-CNN	Faster R-CNN (RPN: resnet50)	Faster R-CNN (RPN: mobile net)
Suggestion box extracting method	Selective search	RPN	RPN
Selective search boxes amount	2000	300	-
Base feature extracting network	VGG16	Resnet50	Mobile net
mAP(%)	85.4	91.1	90.4
Processing speed for each image(ms)	1830	206	59

It can be seen from table 3 that the detection performance of Faster R-CNN on different basic feature extraction networks is very different. Compared with Resnet50, the Mobile-net network is a small convolutional network, which is used as the base feature extraction network of Faster R-CNN, the mAP of the Resnet50 network can basically reach more than 90%, while the mAP of Mobile net is about 90%. However, Mobile-net's image processing speed is obviously about 3 times faster than Resnet50 network. In the actual equipment detection scenario, Resnet50 is still acceptable for each image processing time of about 0.2s. Therefore, considered two

factors of recognition accuracy and processing rate, Mobile net is better than the Resnet50 network. On the other hand, when the Faster R-CNN and Fast R-CNN algorithms are compared, the results show that the average recognition accuracy of the Faster R-CNN algorithm is significantly better. For Fast R-CNN algorithm, as for image processing speed, Faster R-CNN is 8 times faster than Fast R-CNN algorithm. Therefore, Faster R-CNN has obvious advantages in the scene of equipment detection in power room.

4.3.3. Comparison of performance for different object detection algorithm

We compared different object detection algorithms for equipment detection in power room, and got their mAP and recall curve as performance evaluation, which is shown in Fig 3 and Fig 4.

As shown in Fig. 3 and Fig. 4, compared with Faster R-CNN that hold Mobile Net as base feature extraction network, Faster R-CNN that hold Resnet50 as RPN network got higher value, whether mAP or recall, the reason may be that Resnet50 take more deeper network and take more capable to extract features for equipment images. On the other hand, Faster R-CNN belongs to Two-Stage method, compared with other One-Stage method, such as SSD and YoloV3 in figure above, Faster R-CNN have more smooth recall curve and stable mAP value. Though Faster R-CNN may take more time to recognize images, it takes better performance.

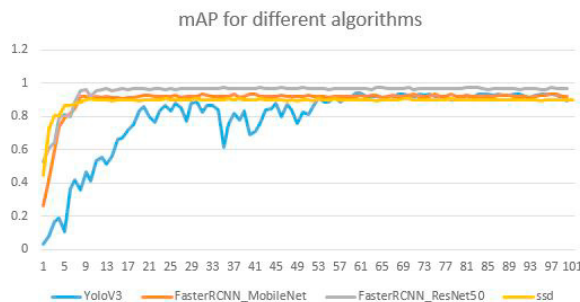


Fig. 3. mAP for different algorithms

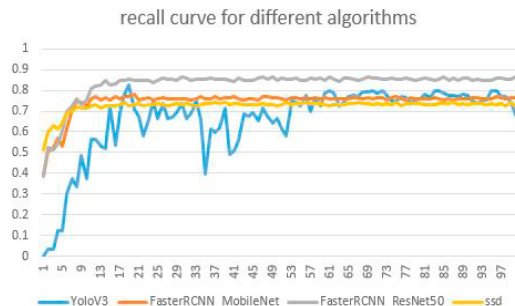


Fig. 4. Recall curve for different algorithms

We applied Faster R-CNN that take Resnet50 as RPN network to the equipment detection tasks, the result is shown in Fig. 5.



Fig. 5. Recognition for equipment detection task

5. Conclusion

This paper proposed a method for equipment detection based on deep learning algorithms. Comparative experiments with other methods show that the method used in this article can achieve better performance in the power room equipment detection task. However, the method in this article is not good for some dense targets, and some targets have been misdetected or missed. Due to the power room Limitations, some small devices are placed compactly, it is difficult to extract an effective feature map, which makes it impossible to effectively optimize the

final detection model. Therefore, in future research, we will focus on solving the detection problem of dense small equipment in the power room to obtain higher accuracy and detection results.

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